EXTENDED ABSTRACT
ACTIVITY TRACKING WITH THE REACHY MINI ROBOT

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Abstract

Juvenile dermatomyositis (JDM) is an autoimmune condition affecting children, characterized by muscle weakness. JDM patients regularly perform physiotherapy exercises assessed by healthcare professionals with the help of Childhood Myositis Assessment Scale (CMAS) scale. Such assessments imply frequent trips to the hospital that can be tiresome and inconvenient for patients and their families. This research explores whether Reachy Mini robot can assist in performing such exercises at home. A prototype was created that uses the robot's camera to track and follow a person, recognize physiotherapy exercises and provide feedback via movements of its head and antennas. The Reachy Mini robot was chosen because it has a camera, the capability to move and computing on board and thus has everything to serve as an exercise companion independently of external processing resources. Development and testing have been conducted both in the simulation environment and on the real robot. Several approaches to pose estimation and activity recognition have been analyzed to find solutions appropriate for the restrictions of the robot hardware. Results show that the robot can conduct real-time person tracking and recognize exercises on its processor. An optimized approach to camera video processing helped to increase person recognition accuracy from 52The paper proves the technical possibility of creating a system that can track patients' exercise progress with the help of the Reachy Mini robot. However, the prototype doesn't perform CMAS scoring and needs further work.

1 Introduction

1.1 Background

The Research Centre for Data Intelligence conducts applied research into, among other themes, smart assistants and human-computer interaction. This project was carried out in the context of the centre's "Measuring at Home with AI" research direction. The long-term motivation for that research is to make exercise monitoring more accessible to children with juvenile dermatomyositis (JDM). These children currently perform exercises related to the Childhood Myositis Assessment Scale (CMAS) under the supervision of specialists at the Wilhelmina Children's Hospital (WKZ). Repeated travel to the hospital is burdensome for children and their families, which creates an opportunity for technologies that can support monitoring at home. Automated home monitoring must do more than record a video. It must detect a person, estimate body movement from visual input, distinguish between activities, and communicate its interpretation clearly to the user. It must also operate under practical conditions, such as changing distances, incomplete views of the body, and limited computing resources. The stakeholders for this feasibility study are the Research Centre for Data Intelligence, the project client and researcher, and the children and families who represent the future users of a home-monitoring system. The project requirements were consolidated in a software requirements specification covering perception, robot interaction, activity recognition, feedback, session management, privacy, accuracy, reliability, usability, and safety. The current project addresses the technical feasibility of such a system; it does not develop or validate a medical assessment instrument.

1.2 Goal

The goal of this project is to investigate whether the Reachy Mini can perform person detection and activity monitoring, including expressive feedback, under the computational constraints of the wireless robot. The resulting artefact is a working prototype that runs in simulation and on the physical robot. It contains a shared camera-to-pose-to-classifier pipeline, temporal feature extraction, activity-recognition models, and robot applications that translate recognized actions into antenna responses. Candidate vision models were benchmarked under the robot's CPU constraints, leading to the selection of MoveNet int8 for on-robot pose estimation. Model-training and export tools were also developed so that the robot can perform inference with NumPy-only model files rather than requiring the full training environment. Recognition models were evaluated on public skeleton datasets with subject or footage-group separation, as well as on robot recordings and an external dataset. Users can also record personalized movements and create their own datasets for future recognition models. The prototype recognizes six classes: none, wave, pushup, squat, clapping, and jumping jacks. Evaluation focuses on technical feasibility using simulation, public skeleton datasets, recorded robot footage, an external test dataset, and different camera framings. The robot's camera should ideally capture the whole person, because waist-up framing removes information that is important for some activities. The prototype is not a CMAS scorer, is not clinically validated, and should not be used as a clinical decision-support system. The central research question is:

> To what extent can the Reachy Mini detect a person, recognize physiotherapy-related exercises, and provide expressive feedback while operating within its computational constraints?

The remainder of this report first discusses the theoretical foundations and related solution space. It then describes the prototype and its evaluation method, presents the results, discusses their implications and limitations, and concludes with directions for future development.

2 Methods

This study employed a custom, iterative methodology to investigate and develop an activity-monitoring system for the Reachy Mini. The approach shares similarities with the Design Science Research (DSR) framework, which is suitable for design-oriented research involving the development and evaluation of an artefact [1]. The approach was selected because the technical feasibility of the system was initially uncertain and several technical choices required investigation before implementation.

The methodology consisted of five phases:

1. Research: identifying stakeholder needs, establishing requirements, and investigating existing techniques.
2. Design: selecting suitable models and defining the technical architecture.
3. Realisation: implementing the activity-recognition pipeline and integrating it with the robot.
4. Testing: evaluating model performance and system behaviour under different conditions.
5. Evaluation: assessing feasibility, identifying limitations, and formulating recommendations.

The phases were not strictly sequential. Research informed design decisions, while implementation and testing were conducted iteratively. Results from experiments were used to refine technical choices and improve the artefact. This approach supported early validation and incremental improvement, which suited the exploratory nature of the project [2]. A traditional waterfall approach was considered less suitable because the technical choices were not fully determined at the start and the project required opportunities to incorporate findings during implementation [3].

Figure 1 illustrates the iterative prototyping approach. Figure 2 shows the activities conducted during the research phase.

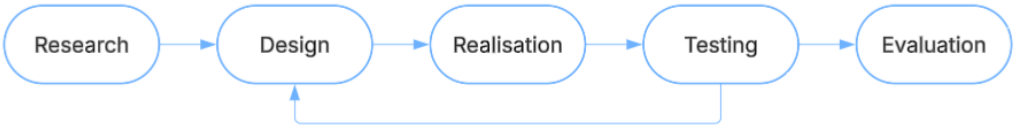


Figure 1: Iterative prototyping approach used in the project.

2.1 Requirements and software requirements specification

The research began with a stakeholder analysis to identify the parties involved and their interests and needs. Functional and non-functional requirements were derived from the project description and meetings with the client. These requirements were consolidated in a software requirements specification (SRS), which served as the reference for design and evaluation.

The functional requirements cover perception and local processing, robot behaviour and interaction, activity and exercise recognition, assessment and feedback, and session management. The non-functional requirements address privacy, recognition accuracy, reliability, usability, and safety. The complete SRS, including all functional and non-functional requirements, is included in Appendix A.

The research was organised around two complementary activities. First, existing techniques for person detection, pose estimation, and activity recognition were investigated. Second, relevant scientific literature, datasets, standards, and guidelines were studied to identify suitable approaches and relevant constraints. The findings were compared against the system requirements and the Reachy Mini's camera and computational capabilities. A detailed hardware and technical capability assessment was conducted to document the robot's hardware architecture, processing capabilities and technical limitations (see Appendix C).

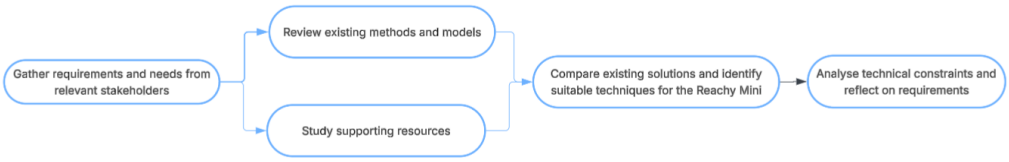


Figure 2: Research activities used to establish requirements and identify suitable techniques for the Reachy Mini.

2.2 Realisation Phase

In the realisation phase, the research was turned into a working prototype. Because the technical choices were not yet fully determined at the start, design proceeded through short build-test cycles in which each design decision was validated on a functional prototype as soon as possible. A simulation-first strategy was chosen because building on the physical robot was not always practical. The code was therefore organised by where it must run. A shared core that is identical in both environments, robot applications, simulation tools (a MuJoCo simulation of the Reachy Mini and a webcam-based simulation on the laptop). All dependencies point only into the core, so that one model file and one feature implementation serve both the simulation and the robot.

2.3 Testing Phase

Testing was not carried out as a single stage after the build. But was integrated into the short design cycles. Each component was checked in simulation first and then run on the physical Reachy Mini robot. In this way, problems that only appear on the real hardware were found and fixed. Pose models were benchmarked on the robot's CPU to determine which estimators are feasible, and the detection rate and keypoint confidence of the chosen pose backend were measured on robot footage and on replayed recordings. In addition to CPU benchmarking, GPU acceleration was investigated to determine whether the Reachy Mini's integrated VideoCore V3D GPU could be used for pose estimation. Additional details on the GPU acceleration experiments and their implementation are provided in Appendix B. For the model evaluation, three data contexts were used deliberately: public datasets as the in-domain measure, recordings made with the robot itself as the realistic "robot in a room" measure, and external datasets as deliberately pessimistic checks of the domain gap. Where a class had not yet been measured on robot footage, this was recorded as not measured rather than extrapolated. Robustness to a partial view of the person was tested by removing the hip, knee and ankle keypoints from the test windows, simulating waist-up framing. The complete applications were run on the robot to verify frame rate, feedback and timing under real conditions.

2.4 Evaluation Phase

In the evaluation phase, the prototype was assessed against the requirements established in the research phase and against the central research question. The evaluation was structured as requirements verification and a reflection on the research process, rather than as a clinical or user validation: the prototype is explicitly not a CMAS scorer, and engagement could not be measured because no user study was conducted. The requirements verification established, per requirement, whether the prototype meets it, or does not meet it. The process reflection evaluated the chosen approach itself: which decisions contributed and which aspects limited the results. The complete requirements verification are provided in Appendix D.

3 Results

3.1 Description of the Artefact

The end product is a software application: a working prototype (an instantiation, in Design Science Research terms) that turns the Reachy Mini into a CPU-only activity monitor that answers what it sees with its own body. It consists of three parts: a layered software architecture that runs the same recognition code in simulation and on the robot, recognition models exported as NumPy-only files, and robot applications built on both. It addresses the requirements of the software requirements specification (SRS) attached to this article (Section 3.1).

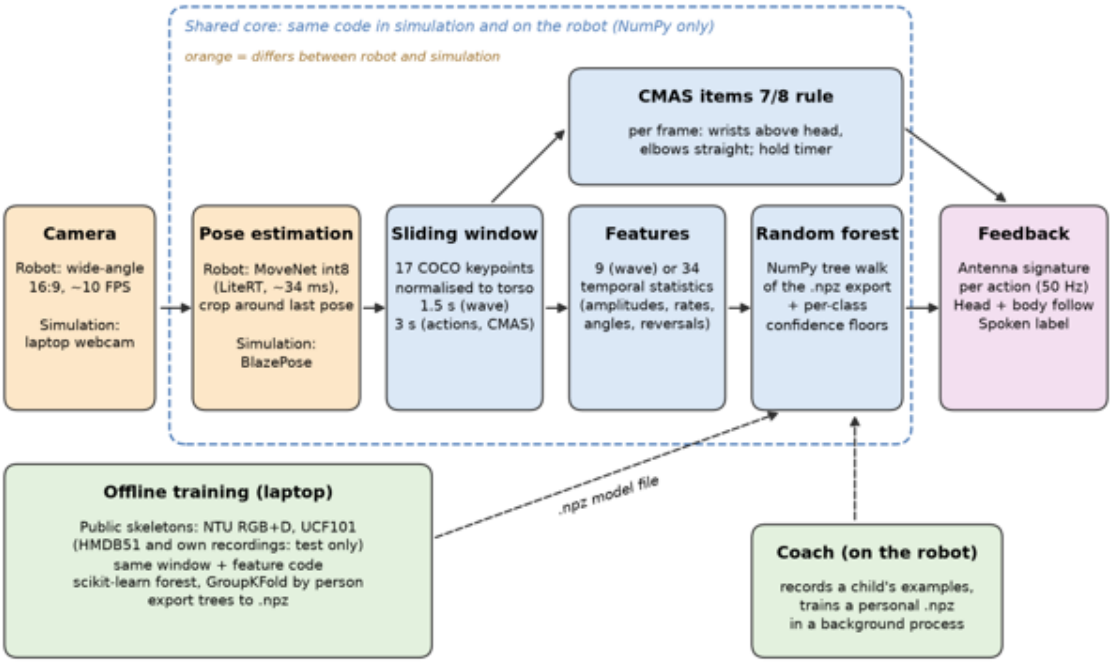


Figure 3: Pipeline Architecture.

Figure 3 represents the artefact's pipeline. Everything inside the dashed line is the same code in simulation and on the robot; only the camera and the pose estimator differ. Models are trained offline, or by the coach on the robot, and loaded as `.npz` files.

The pipeline works in such way that a camera frame goes to a pose estimator that returns 17 COCO keypoints: MoveNet Lightning int8 on the robot, BlazePose on the laptop webcam in simulation. The keypoints enter a sliding window, are normalised to the torso and summarised as temporal features. A random forest, evaluated by a NumPy-only tree walk, classifies every window, and a per-class confidence floor turns uncertain windows into *none*. Each detection drives the feedback: a distinct antenna signature per action, rendered at 50 Hz independently of the 10 FPS vision loop, head and body following that keeps the person in view, and an optional spoken label.

As for architecture and organization, the code is organised by where it must run: a shared core (pose backends, features, models, feedback), robot applications, simulation tools for MuJoCo and a webcam, and training tools, which are the only part that needs scikit-learn. Dependencies point only into the core, so one model file and one feature implementation serve both environments, and training and inference cannot drift apart.

For the final application, the robot runs action recognition (the six classes, or the CMAS items of Section 4.6), wave recognition, head and body tracking, and a coach for children. The coach guides a child through about 4.5 minutes of recording (squats, arm raises, side bends), trains a personal model on the robot's own CPU while the head keeps following, and then runs a workout with that model. A preflight tool brings the robot into a known state before every application.

Table 1 links the main decisions to the requirement or insight behind them.

3.2 Pose-Model Performance

The benchmark compared BlazePose, MoViNet-A0, TSM-MobileNetV2, ViTPose and MoveNet as candidates for the perception pipeline. MoveNet Lightning int8 reached

Table 1: Factors Influencing the System Design.

Origin	Decision
Requirement: local processing on a CPU without GPU	MoveNet int8 for pose; forests exported to NumPy instead of a deep skeleton network
Requirement: privacy of child users	Only keypoints are stored; children's recordings and models stay on the robot
Grounding: windowed data leaks between splits	Every score from GroupKFold by person
Grounding: public datasets may not transfer to a room	Robot and room recordings kept as external test sets only
Design cycle: BlazePose cannot run on the robot's CPU (its binary needs AES instructions)	Pose backend made swappable behind one interface
Design cycle: a table-mounted camera often sees the person waist-up	Training windows also built with hips and legs removed
Design cycle: letterboxing the 16:9 wide-angle frame left the person a few dozen pixels tall	Pose crop centred on the previous pose; person detected in 72% of frames instead of 52%
Design cycle: scikit-learn inference cost ~33 ms per window on the robot	Trees exported to `.npz` and walked in NumPy
Design cycle: the robot often boots asleep, motors off, camera released	Preflight with automatic fix

23.80 FPS in the inference benchmark, with a mean pose-inference time of approximately 34 ms on the Reachy Mini; in the complete application the camera, at about 10 FPS, sets the rate (Section 4.5). This was the most suitable result for the CPU-only target, and the model produces the COCO-17 keypoints the recognition code expects. Cropping each frame around the previous pose, instead of letterboxing the whole wide-angle frame, raised the share of robot frames with a detected person from 52% to 72%. On a replay of a laptop recording with the robot's aspect ratio, it raised the mean keypoint confidence from 0.48 to 0.68.

3.3 GPU Acceleration Experiments

The two GPU implementations used LiteRT and WebGPU. In MoveNet, an ARM64 LiteRT runner could be created, while in BlazePose a native C++ library as well as a GPU inspector could be built. The two implementations could be tested in runtime, but both failed at the GPU initialization step. After further investigations, it was found out that there was a compatibility problem with the prebuilt LiteRT WebGPU accelerator. It included AES instructions, which were not supported by the Reachy Mini's Cortex-A72 CPU and thus caused an illegal instruction problem. The problem appeared to be independent of the camera and model inference, meaning that it was caused by the GPU runtime and not by the models themselves. Neither of the implementations could thus reach the GPU inference step. The full code of the GPU implementations are available in the /repos section of the repository. Build results are included in Appendix B.

3.4 CMAS Items

In CMAS mode the prototype recognises that four CMAS items were performed; it does not score them. Items 12 (chair rise) and 14 (pick-up) are classified by a three-class forest (none, chair rise, pick-up) trained on NTU RGB+D alone (actions "standing up" and "pick up"). Items 7 and 8 (arms straight above the head, and for how long) use a geometric rule on single frames, because a held posture has no movement for a window to learn. Item 13 (stool step) was left out: no public dataset contains it and none was recorded.

3.5 Pose-model performance

The benchmark compared BlazePose, MoViNet-A0, TSM-MobileNetV2, ViTPose and MoveNet as candidate components for the perception pipeline. MoveNet Lightning int8 achieved an average of 23.80 FPS, with a mean pose-inference time of approximately 34 ms on the Reachy Mini. This was the most suitable result for the project's CPU-only deployment target. The model also produces the COCO-17 keypoints expected by the temporal feature and recognition code.

The comparison also exposed deployment constraints. BlazePose and ViTPose are useful comparison models, but the robot cannot use the available BlazePose MediaPipe binary because it requires AES instructions unavailable on the Reachy's Cortex-A72; ViTPose's detector-and-pose pipeline is intended for the laptop or simulation environment. MoViNet-A0 and TSM-MobileNetV2 were considered for temporal classification rather than selected as replacements for the keypoint-based pipeline. Consequently, MoveNet int8 was used for robot inference, while the shared downstream representation remained model-independent.

3.6 Wave recognition

On two recorded sessions with our own robot (416 windows), the rule-based detector reached F1 0.811 (precision 1.000, recall 0.682). The NTU-trained forest, which never saw these recordings, reached F1 0.951 and outperformed a forest trained on the sessions themselves (F1 0.884, historical only). On unseen NTU subjects, where the negatives include deliberately confusable gestures (salute, selfie and brushing hair), F1 was 0.556. On HMDB51 movie clips it was 0.065, a deliberately pessimistic external check.

3.7 Six-class action recognition

Table 2: Out-of-fold F1 on held-out subjects or footage groups. Waist-up framing is simulated by removing the hip, knee and ankle keypoints.

Class	Full body	Waist-up
none	0.902	0.855
wave	0.918	0.873
squat	0.922	0.803
clapping	0.898	0.899
jumping_jacks	0.887	0.861
pushup	0.381	0.137

These are in-domain scores. A source confound inflates wave and clapping, both taken from the studio-recorded NTU corpus. On our robot recordings wave drops to 0.581, the figure to quote for a robot in a room, and clapping has not yet been measured on robot footage. Squats are almost never false positives (precision 0.998) but miss about one repetition in seven (one in three waist-up). Push-ups (19 training groups, per-fold F1 0.06–0.71) are insufficiently measured and are not demonstrated. Without crop augmentation, waist-up F1 for wave, squat and clapping fell to 0.000.

4 Discussion

Several choices contributed positively to the results. Training on skeletons from public datasets gave the project a fast start: working recognition models existed before any data of our own had been recorded, and no video had to be processed. That head start had a limit, however. The public datasets did not cover every exercise the project needed, and they did not show a person the way the robot sees one, so we later had to record our own sessions to train and test the models that mattered most. A second positive choice was to run the same code in simulation and on the robot. Features, models and feedback were developed and checked on a laptop, then deployed to the robot unchanged, which kept the two environments from drifting apart. Finally, short design cycles on the physical robot exposed problems that no laptop test would have shown. They revealed that BlazePose cannot run on the robot's processor, and that letterboxing the wide-angle image left the person too small to detect reliably. Each problem was found and fixed within days, and the crop around the previous pose raised the share of frames with a detected person from 52 percent to 72 percent.

Other aspects limited the results. The robot's camera does not match the image quality of a laptop or phone camera. Its wide-angle view leaves a person only a small part of the image, so keypoints are less certain than in simulation. The CPU, without a GPU, also restricted the choice to a lightweight pose model. LiteRT and WebGPU setups have been tested for GPU acceleration, but both were built to ARM64 level and had problems initializing the GPU because of unsupported AES instructions in the accelerator. A further limit was the difference between the training videos and real use. The public datasets were recorded in studios or taken from films, with other backgrounds, camera angles and sometimes several people in view. The robot, in contrast, sees one person in a home room from table height. This difference lowered recognition quality outside the datasets: waves, for example, scored F1 0.918 on dataset subjects but 0.581 on our robot recordings. The robot was also not always available to the team. The simulation largely mitigated this risk, because most of the development could continue without it. Lastly, engagement could not be measured, because the prototype could not be tested with real users. Whether children find the robot an engaging exercise buddy therefore remains an open question.

5 Conclusion

The goal was to find out whether the Reachy Mini can detect a person, recognise their activity and give expressive feedback within the limits of its own CPU. GPU acceleration was also investigated through two LiteRT/WebGPU implementations using MoveNet and BlazePose. Although both implementations were successfully prepared for ARM64, neither reached GPU inference because the shared WebGPU accelerator contained AES instructions unsupported by the Reachy Mini's Cortex-A72 processor. This established a concrete compatibility limitation and provides a starting point for future work. Rebuilding the accelerator for the target processor is the most direct next step. The complete pipeline runs on the robot's Raspberry Pi CM4 without a GPU or a laptop. Pose estimation takes about 34 ms per frame, features and forest a few milliseconds, and the camera's 10 FPS, not the computation, sets the pace. The same code and models run in simulation, so they are developed on a laptop and deployed unchanged. Recognised actions are answered in real time with antenna signatures and head and body following.

The answer to "how well" is more limited. In live use, wave, clapping and jumping jacks were reliable enough to demonstrate, and squats were recognised with about one repetition in seven missed. Push-ups were not. Scores on public datasets overstate what happens in a room: the six-class model recognises waves with $F_1 = 0.918$ on dataset subjects but $F_1 = 0.581$ on our robot recordings. For the CMAS items, pick-up held up in a room ($F_1 = 0.933$). Chair rise and arms above the head were recognised live on the robot, but they have not yet been measured on a correctly performed room recording. The robot's position also matters: it must see the whole person, and a waist-up view costs recognition.

The prototype, its models, the training and export tools, and the coach were realised. A CMAS score, the stool-step item and clinical validation were not; they were outside the scope. The "engaging" part of the research question was not evaluated, since no user study was done. The prototype is therefore a research platform: it shows that monitoring at home with this robot is technically feasible, but it cannot yet be used to monitor children with JDM.

That said, there are four recommended next steps:

1. Record labelled sessions in real rooms through the robot's own camera, following the CMAS instructions, and use them as the reference for room performance.
2. Collect data for the classes that lack it: push-ups, clapping on robot footage, and the stool step.
3. Run a small user study with children and families to measure engagement, which the research question assumes, but this project did not test.
4. If CMAS scoring becomes the goal, define the scoring criteria with the WKZ specialists and validate the system against their scores.

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Appendices

A System Requirements Specification

M-ADSAI-ZM002-26

System Requirements Specification

Caio De Alcantara Santos (2604500), Cas Limpens (2400410), Donné Deguelle (2301667) and Gustavo Gonçalves Da Costa (2604824).

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Introduction

Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for a system that uses the Reachy Mini robot and AI-based movement detection to monitor and evaluate human movements during physiotherapy exercises at home. The project investigates whether the Reachy Mini can be used as an interactive and motivating exercise buddy. The system should be able to detect and potentially score common human movements and provide feedback or encouragement to the user. The SRS defines the functional and non-functional requirements of the proposed system and serves as a basis for its development and evaluation.

Scope

The scope of this project is to investigate the capabilities of the Reachy Mini for person detection and activity monitoring. The system will focus on detecting common movements such as walking, running, sitting, standing, and other relevant physiotherapy movements. The project will investigate whether existing AI and pose-estimation models can identify the movement being performed, determine whether it is performed correctly, and potentially provide a score. The Reachy Mini may also provide interactive feedback through movements, sounds, or voice to increase motivation. The project does not aim to solve a specific medical case or provide a complete medical monitoring solution. Instead, the focus is on investigating whether the Reachy Mini can identify and score general human movements and function as a motivating device for home-based physiotherapy.

Intended Users

The primary intended users are patients performing physiotherapy exercises at home. The system is intended to support users who perform exercises independently by providing motivation and potentially feedback on their movements. The Reachy Mini is intended to act as an interactive exercise buddy that can react to the user's movements and provide encouragement or feedback. The system may also be relevant to physiotherapists and healthcare professionals, who could use the Reachy Mini as a replacement assistant for appointments and scoring, to support the monitoring of patients' exercises. The original project document specifically mentions children with JDM (Juvenile Dermatomyositis) performing exercises that are assessed using the CMAS (Childhood Myositis Assessment Scale), but this project broadens the current project goal toward general physiotherapy movements and investigating the Reachy Mini as a motivating device.

System Overview

Product Description

The product is a movement detection system that uses the Reachy Mini robot's capabilities and AI algorithms to detect and score JDM patients movement during physiotherapy exercises at home. The system aims to provide motivation to patients. To support exercise performance.

Users and Stakeholders

The intended users of the Reachy Mini are JDM patients. But also, healthcare professionals who assess these patients. The robot could help assess the patients, without the patients needing to travel to the hospital as often.

Our client Maarten Vaessen who has tasked us with this project, is a senior researcher in AI for healthcare at Zuyd research center data intelligence (Zuyd University of Applied Sciences, n.d.).

Assumptions and Dependencies

The Reachy Mini can be programmed with the use of the Reachy Mini python SDK. Which is supported on Windows, MacOS and Linux. The python SDK requires python version 3.10-3.12 and the latest version of git, including git LFS (Huggingface, n.d.). It is assumed that the client will provide the Reachy Mini DIY kit.

Functional Requirements

The functional requirements describe what the system has to do. Each requirement has a unique identifier, is written using the modal verb *shall*, and applies to a single user in front of a single Reachy Mini during an exercise session.

Several requirements refer to *supported activities*. This means the closed set of movements the system is able to recognise: standing, sitting, walking, running, arm raise and sit-to-stand. The set is deliberately small, because the aim of the project is to find out whether recognition of this kind is feasible on the Reachy Mini, not to cover the full range of physiotherapy exercises.

Perception and local processing

Before an activity can be recognized, the system must detect and track the user. The activity recognition approach may either use an explicit skeletal or pose representation or infer activities directly from image or video data. Both approaches are therefore considered within the scope of this research. FR-02 additionally requires that this happens locally, which is the constraint that gives the project its research character and from which the privacy property in NFR-01 follows.

Table 1 – Perception and local processing functional requirements

ID	Title	Requirement
FR-01	Person Detection	The system shall detect a person within the Reachy Mini's camera view within 5 seconds after the person enters the camera view during an exercise session.
FR-02	Local Processing	The system shall process captured camera data entirely on the Reachy Mini or the connected local computer, without requiring external processing for core activity-tracking functionality.
FR-03	Person Tracking	The system shall track the detected person continuously while the person remains visible within the camera's field of view during an exercise session.
FR-05	Pose Estimation	The system shall provide the input representation required by the selected activity recognition approach, which may be an explicit pose/skeletal representation or image/video data.
FR-18	User Detection Loss	The system shall detect that the user is no longer visible to the camera within 2 seconds after the user leaves the camera's field of view during an exercise session.

Robot behaviour and interaction

The robot moves for two different reasons. In FR-04 the head moves in order to keep the user in view, which is a perception function. In FR-13 and FR-14 the head and antennas move in order to communicate with the user, which is an interaction function. How these two influence each other is one of the questions the project sets out to answer.

Table 2 – Robot behaviour and interaction functional requirements

ID	Title	Requirement

FR-04	User Following	The system shall move the Reachy Mini's head to follow the detected user while the user remains within the camera's field of view during an exercise session.
FR-12	Audio Feedback	The system shall provide audio feedback through the Reachy Mini's speaker whenever an audio feedback response is triggered.
FR-13	Head Movement Feedback	The system shall use movements of the Reachy Mini's head to provide feedback to the user whenever a head-movement feedback response is triggered.
FR-14	Antenna Movement Feedback	The system shall use movements of the Reachy Mini's antennas to provide feedback to the user whenever an antenna-movement feedback response is triggered.

Activity and exercise recognition

Three categories are distinguished because they place different demands on the recognition approach. Static positions can be classified from a single pose, dynamic movements require a sequence of poses over time, and physiotherapy exercises additionally have a defined correct execution.

Table 3 – Activity and exercise recognition functional requirements

ID	Title	Requirement
FR-06	Static Movement Recogntion	The system shall recognise all static positions listed as supported activities during an exercise session.
FR-07	Dynamic Movement Recognition	The system shall recognise all dynamic movements listed as supported activities during an exercise session.
FR-08	Physiotherapy Exercise Recognition	The system shall recognise all physiotherapy exercises listed as supported exercises during an exercise session.

Assessment and feedback

Recognising a movement is not enough on its own, since the robot is meant to act as an exercise buddy. FR-09 and FR-10 cover the judgement about the execution, and FR-11 covers the decision to respond. The assessment is explicitly non-clinical: the movement criteria describe geometric properties such as joint angles and symmetry, and do not constitute a medical score.

Table 4 – Assesment and feedback functional requirements

ID	Title	Requirement
FR-09	Movement Assessment	The system shall assess the execution of supported physiotherapy exercises using the predefined movement criteria for each exercise.
FR-10	Repetition Detection	The system shall detect completed repetitions according to the predefined repetition criteria during an exercise session.

Session management

An exercise session is the unit of operation of the system: the user starts it, the system observes continuously during it, and the results are reported at the end.

Table 5 – Session management functional requirements

ID	Title	Requirement
FR-11	Session Management	The system shall allow the user to start and end an exercise session.
FR-12	Exercise Monitoring	The system shall monitor the user's movements continuously throughout an active exercise session.
FR-13	Exercise Result	The system shall provide the user with the detected exercise results and assessment results at the end of an exercise session.

Non-Functional Requirements

The non-functional requirements describe the qualities the system has to exhibit rather than the functions it has to perform. Each is expressed as a measurable target, so that the outcome of the investigation can be reported objectively.

Privacy is a driving constraint rather than a secondary concern, because the system observes a patient exercising at home. Accuracy determines whether the system is useful at all, since feedback based on a misrecognised movement is worse than no feedback; the threshold of 80% demonstrates technical feasibility within the scope of this project and is not a level that would be sufficient for clinical use. Reliability and usability follow from the fact that the system is operated by a patient without technical support present, and safety follows from the robot moving in the physical presence of a user who may be a child.

Table 6 – Non-Functional Requirements

ID	Title	Requirement
NFR-01	Privacy	The system shall ensure that captured video and session data remain on the Reachy Mini and/or the connected local computer for the entire duration of system operation and shall not be transmitted to external systems.
NFR-02	Recognition Accuracy	The system shall provide activity recognition for the supported activities when evaluated using the predefined test dataset. The recognition performance shall be measured and reported.
NFR-03	Reliability	The system shall complete an exercise session without requiring an application restart under normal operating conditions.
NFR-04	Usability	The system shall allow first-time users to start an exercise session without assistance from the development team.
NFR-05	Safety	The system shall not perform unexpected or uncontrolled robot movements during normal operation.

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B GPU Acceleration Research

M-ADSAI-ZM002-26

Reachy Mini: GPU Acceleration Research

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Introduction

Purpose

The purpose of this research is to investigate the feasibility of using the integrated GPU of the Reachy Mini to accelerate AI inference, with a particular focus on human pose estimation. This research builds upon the earlier Reachy Mini: Hardware & Technical Capability Assessment (De Alcantara Santos, Limpens, Deguelle, & Gonçalves Da Costa, 2026), which examined the robot's hardware specifications, computing capabilities, available software interfaces and suitability for running AI models locally. This research builds upon those findings by focusing specifically on GPU acceleration. Different frameworks and implementations are investigated to determine whether the Broadcom VideoCore V3D GPU can be used for machine learning inference and which technical limitations prevent successful execution.

Scope

The scope of this research is GPU acceleration of human pose estimation on the Reachy Mini. The goal of the research is not to build a complete GPU-accelerated application. It is rather an attempt to document the process of development, establish how far each implementation goes and understand what needs to be done for successful GPU inference in the future.

Research Questions

The following research questions define the focus of this research.

Main Question

“To what extent is it possible to accelerate human pose estimation on the Reachy Mini using its integrated Broadcom VideoCore V3D GPU, and what technical limitations prevent successful GPU inference?”

Subquestions

1. What are the hardware capabilities and limitations of the Reachy Mini's Raspberry Pi Compute Module 4, and which GPU acceleration frameworks are suitable for this hardware?
2. How can LiteRT and WebGPU be configured and cross-compiled for ARM64 to run MoveNet and BlazePose on the Reachy Mini?
3. What are the differences between the MoveNet LiteRT/WebGPU implementation and the native C++ BlazePose LiteRT/WebGPU implementation?
4. What caused the AES-related runtime failure in both LiteRT/WebGPU implementations, and how was the problem identified?
5. What technical improvements and development steps are required to achieve successful GPU inference on the Reachy Mini in future work?

Methodology

The research has been conducted by iterative implementation, cross-compilation and testing on the Reachy Mini. The GPU applications have been prepared initially on the Ubuntu x86_64 development machine using ARM64 cross-compilation toolchain. Then the resulting binaries and libraries have been tested on the Reachy Mini. In case of failure of GPU initialization additional diagnostics have been performed to find out the processor capabilities, binary instructions, shared library dependencies and runtime loading.

Hardware and Technical Context

The capabilities of the Reachy Mini's hardware were analyzed thoroughly in a previous work entitled Reachy Mini: Hardware & Technical Capability Assessment (De Alcantara Santos, Limpens, Deguelle, & Gonçalves Da Costa, 2026). The robot uses a Raspberry Pi Compute Module 4 with a Broadcom BCM2711 quad-core Cortex-A72 processor, 4 GB of RAM and an integrated Broadcom VideoCore VI GPU (Raspberry Pi, sd). The robot also has IMX708 camera, which serves as a source of video input needed for pose estimation. The prior work also describes other components that can be found on the robot such as sensors, cameras, software development kits, APIs, and networking capabilities.

In this work, the GPU architecture and its software compatibility are the crucial part. The VideoCore GPU can be accessed through Mesa graphics stack and V3D driver(Mesa Project, sd). As the hardware doesn't support NVIDIA CUDA, the alternatives like LiteRT and WebGPU have been analyzed. The following sections describe these technologies and the problems arising from the attempt to use them on the Reachy Mini.

GPU Implementations

GPU Implementation 1: MoveNet with LiteRT and WebGPU

The first dedicated GPU approach focused on MoveNet Lightning INT8 (STMicroelectronics, sd). It was supposed to employ the same model as the future working proof of concept for CPU but with the replacement of the XNNPACK CPU backend with GPU acceleration of the model provided by LiteRT. The latter is Google's runtime for on-device machine learning inference with the APIs for loading models and choosing hardware accelerators supported (Google AI Edge, sd). The envisioned architecture involved feeding the camera into the Python application, which was supposed to preprocess the frames into a 192 × 192 RGB tensor, pass it to the native ARM64 LiteRT runner and get the keypoints to draw a skeleton on top of the camera image.



Figure 1 Intended architecture of the MoveNet LiteRT/WebGPU implementation on the Reachy Mini

The LiteRT runtime was built cross-platform on an Ubuntu 24.04 x86_64 development machine using the aarch64-linux-gnu toolchain. In particular, an initial problem arose due to the attempt to run the ARM64 version of the FlatBuffers compiler (flatc) on the x86_64 host platform. It led to an Exec format error. The problem was solved by compiling flatc as a native x86_64 host tool and configuring the build to use that one. With this fix in place, the ARM64 runner was successfully built. It resulted in the files including the LiteRT runner, libLiteRt.so, the prebuilt libLiteRtWebGpuAccelerator.so and the MoveNet INT8 model.

The Python application was then hooked up to the runner. GPU acceleration was explicitly requested in the accelerator configuration including --accelerator=gpu, --gpu_backend=webgpu and --require_registered_accelerator=gpu options. The last one was crucial since it avoided silent falling back to the CPU.

The main problem was GPU initialization. The failure occurred during the first GPU registration check when the MoveNetLiteRT initialization routine was invoked. Specifically, the ARM64 LiteRT runner tried to load the prebuilt WebGPU accelerator library, which was compiled with the AES instructions not supported by the Reachy Mini's Cortex-A72 processor. It caused SIGILL ("illegal instruction" error), leading to failure of the initialization routine. Since the application checks that before connecting to the Reachy Mini camera, the camera processing, image preprocessing and MoveNet inference were not even attempted.



Figure 2 LiteRT WebGPU initialization failure during accelerator registration

GPU Implementation 2: Native BlazePose with LiteRT and WebGPU

The second GPU implementation used the BlazePose Full framework and followed another strategy compared to the MoveNet implementation. Instead of using an existing command-line runner, a custom native C++ inference pipeline was implemented to have greater control over the model execution. The implementation used the original MediaPipe Pose Landmarker Full model bundle, which consisted of two separate TFLite models: a pose detector and a pose landmark model (Google AI Edge, sd). The detector locates a person and defines a region of interest (ROI), and then the landmark model infers the body pose. The original pose_landmarker_full.task bundle was separated into pose_detector.tflite and pose_landmarks_detector.tflite. Both models were loaded separately using LiteRT and set up to run with the GPU backend. The intended processing pipeline was to take a frame from the Reachy Mini camera, transfer it from Python to the native C++ library through ctypes, run the detector on the GPU, calculate the ROI in C++ and then run the landmark model on the GPU. After that, the resulting landmarks should be transferred back to Python to visualize the skeleton and run benchmarking.



Figure 3 Native BlazePose inference architecture using two LiteRT GPU models and a custom C++ processing pipeline

The native implementation was split into the Python interface and the C++ processing part. The Python adapter, which was implemented in gpu_models/__init__.py, loads libblazepose_gpu.so through ctypes and invokes the exported C functions. The C++ library provides the API for creating the GPU pipeline, processing an image, getting the error information and destroying the allocated resources. This allowed to keep using the existing Python benchmark while providing the new implementation, without modifying the original CPU-based one. At the initialization stage, the C++ code creates a LiteRT environment and the GPU execution options. It explicitly enables the GPU acceleration using SetHardwareAccelerators(HwAccelerators::kGpu). Then both the detector and landmark models are loaded using CompiledModel::Create(). Their input and output tensor buffers are created once and stored in the native BlazePoseGpu object. This means that the models and their execution contexts are supposed to remain loaded between camera frames, and not to be recreated at every inference.

At the inference stage, Python converts each camera frame from BGR to RGB and sends the resulted image data to the native library. In C++, the image is resized to 224 × 224 pixels using the letterboxing technique and normalized to the range [-1, 1]. Then the detector runs through LiteRT with the GPU backend. The output includes detection boxes and confidence scores, which are then copied to the C++ memory and further processed using the anchor decoding and NMS. The top-scoring detection is used to calculate the ROI, which consists of the position, scale and rotation of it. Then the region of the image is cropped and rotated to a 256 × 256 input of the landmark model. This second model also runs through LiteRT GPU backend. The output is read back to the C++ memory, and then the coordinates of the first 33 public pose landmarks are converted back to the original image coordinates. Finally, the native library sends the coordinates of the landmarks along with the confidence values to Python, where the skeleton is

drawn on the camera image. The benchmark also collects inference latency, FPS, CPU and RAM usage.

aarch64-linux-gnu-g++ cross-compiler. CMake was configured with CMAKE_SYSTEM_PROCESSOR=aarch64 and the corresponding ARM64 compiler. The compilation depended on the existing LiteRT source and build environment located under ~/LiteRT, including the already built ARM64 LiteRT libraries. An initial error during compilation was related to the aggregate initializer used for initializing the BlazePoseGpu structure with the wrong number of fields. The structure needed an explicit constructor, which accepts a LiteRT environment and two compiled models. This was fixed by using the constructor directly and moving the model objects with std::move(). After this modification, the compilation finished successfully producing libblazepose_gpu.so and blazepose_gpu_inspector. Both binaries were later verified using the Linux file command and proved to be compiled for ARM aarch64.

After the successful cross-compilation, the next step was to test the GPU implementation directly on the Reachy Mini. For this purpose, a separate executable blazepose_gpu_inspector was built. It initializes the LiteRT environment, requests the GPU acceleration, tries to create a CompiledModel for each of the two BlazePose models and checks whether the tensor buffers use the WebGPU memory. The Python benchmark uses almost the same initialization sequence. First, it loads the native library through ctypes and calls blazepose_gpu_create(). This function creates the LiteRT environment, sets up the GPU execution and tries to initialize the detector and landmark models. Importantly, this initialization happens before the benchmark starts its camera warm-up and frame-processing loop. At the first GPU model initialization, LiteRT tries to load the pre-built libLiteRtWebGpuAccelerator.so library. The same AES-related illegal-instruction error occurred, which was observed in the MoveNet implementation. The Cortex-A72 processor of the Reachy Mini does not support these instructions of this binary, so the process terminates before the first CompiledModel can be created successfully. As a consequence, neither detector nor landmark models reach the inference stage, and no GPU tensor buffers or pose landmarks are generated. The successful ARM64 compilation thus proves that the custom application and its native binaries can be built for the Reachy Mini, but it doesn't prove that the whole GPU pipeline can run on it.



Figure 4 BlazePose LiteRT/WebGPU initialization failure caused by unsupported AES instructions on the Reachy Mini

Shared LiteRT WebGPU Runtime Failure

One of the important discoveries made in this case was that the implementation of MoveNet and BlazePose failed not due to two distinct issues related to the model. In both implementations, the pre-built LiteRT WebGPU accelerator was used - libLiteRtWebGpuAccelerator.so. This was checked by comparing the SHA-256 hashes of the accelerator files in the project, and the hashes matched, meaning that the two implementations have one and the same dependency runtime. The capabilities of the Reachy Mini processor were determined with /proc/cpuinfo command, and the output showed no support of AES and PMULL

instruction sets. Using objdump, it was seen that the library had the instructions such as aesd, aese, aesmc and aesimc, which require certain processor features absent in the target CPU architecture. Although the library is compiled for ARM64, its instruction requirements are not compatible with the specific Cortex-A72 configuration.

This was confirmed by a more direct Python ctypes.CDLL() test - the test tried loading the WebGPU library without initializing the camera, loading the model and conducting inference, but resulted in the same fatal error. Checking the WebGPU library with ldd showed that the standard dependencies of a shared library were correctly loaded, but the main libLiteRt.so library did not exhibit the problem with the AES instruction. This way, the incompatibility was isolated to the supplied WebGPU accelerator.

This is why the problem was not fixed with a change of the Python application or rebuilding the native runner, because the problem with the accelerator was due to the fact that it was supplied as a precompiled library. The LiteRT and WebGPU implementations therefore halted at the same initialization point. The next step would be to rebuild or replace the accelerator and its dependencies compiled with appropriate compiler settings for Cortex-A72. Still, even after successful library loading, WebGPU device selection, model loading and inference will have to be checked.

Future Development and Recommendations

In order to move forward with the LiteRT/WebGPU projects, we propose that the WebGPU accelerator and its dependant components be recompiled for the Cortex-A72. This way, all the necessary ARM64 components from the current implementation would be used, while only the pre-built accelerator would need to be reworked. Firstly, the version of the LiteRT source code needs to be found, then the accelerator should be recompiled under the right ARMv8 settings. The resulting binaries need to be checked for unsupported AES instructions (this can be done using objdump). It would also be helpful to check whether the accelerator works independently before trying to run a full pose estimation app using it. For future developers, it would be useful to make sure that the library loads, that the GPU accelerator is registered by LiteRT and that at least a simple model can be initialized and executed. This way, problems associated with model execution and camera initialization would not be confused with each other.

Conclusion

The greatest success was reached while working on the two LiteRT/WebGPU approaches. In particular, the MoveNet approach succeeded in building an ARM64 LiteRT runner and including it in a Python project. The second approach managed to go further in terms of native implementation and created a custom inference pipeline in C++, ARM64 shared library and GPU Inspector. At the same time, both LiteRT/WebGPU approaches stopped at the GPU initialization stage since they used the same prebuilt WebGPU accelerator which had instructions supported by the AES algorithm which is incompatible with the Cortex-A72 processor of the Reachy robot. It was proved by performing CPU feature inspection, analyzing the binary code and an additional library loading test. Thus, the primary output of the current study is a clear diagnosis of the LiteRT/WebGPU incompatibility issue and a technical base for future research. Modification of the WebGPU accelerator with appropriate compiler options is the most promising way to go forward.

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C Hardware and Technical Capability Assessment

M-ADSAI-ZM002-26

Reachy Mini: Hardware & Technical Capability Assessment

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Introduction

Purpose

The purpose of this research is to investigate the hardware and technical capabilities of the Reachy Mini and determine whether the robot is suitable for our intended application. The research focuses on the processing capabilities of the robot itself, the available sensors and interaction hardware, the available software interfaces, and the connectivity options required to operate the robot. A specific focus of this research is determining whether AI-based pose estimation and movement-recognition models can be executed directly on the Reachy Mini's internal hardware. If the available hardware is insufficient for the required AI processing, this research will investigate the feasibility of using an external computer or other hardware for computationally intensive tasks. The results of this research will allow the team to make an informed decision about which functionality can be performed directly on the Reachy Mini and which functionality may require additional hardware or infrastructure.

Intended Use Case

The intended use case is to use the Reachy Mini as an interactive and motivating device for patients performing physiotherapy exercises at home. The robot could support the patient by providing encouragement, feedback and interactive responses during exercises. It may also react to the patient's movements and use its physical movements, audio output and other interaction capabilities to make the exercises more engaging. AI-based pose estimation and movement recognition are expected to play an important role in this use case. The system should be able to identify which movement the patient is performing, determine whether the movement is performed correctly and, in a later stage, provide a score or other form of feedback.

Scope

The following areas will be investigated:

- The Reachy Mini's internal hardware and specifications.
- Camera capabilities and software accessibility.
- Microphone, speaker and audio capabilities.
- Motors, movement capabilities and control interfaces.
- Other relevant sensors.
- Operating system, SDK and available APIs.
- Processing capabilities for computer vision and AI.
- Feasibility of running pose-estimation and movement-recognition models locally.
- Options for external AI processing.
- Networking and connectivity options.
- Network requirements for the intended use case.
- Additional equipment required for testing.
- Technical limitations and potential risks.

The research does not aim to develop the final application. Instead, it focuses on establishing whether the Reachy Mini provides the technical capabilities and infrastructure required for further development.

Research Questions

The following research questions define the focus of this research.

Main Question

- *To what extent can the Reachy Mini detect a person, recognize physiotherapy-related exercises, and provide expressive feedback while operating within its computational constraints?*

Subquestions

1. *What hardware and technical specifications does the Reachy Mini have?*
2. *What functions and interfaces are available for the camera, audio, sensors and robot movement?*
3. *What software, SDKs, APIs and libraries are available for developing applications for the Reachy Mini?*
4. *What computing capabilities does the Reachy Mini have for running computer vision and AI models locally?*
5. *Can suitable pose-estimation and movement-recognition models run locally on the Reachy Mini, and what limitations should be expected?*
6. *If local processing is insufficient, how feasible is it to use an external computer for AI processing?*
7. *What connectivity and network setup is required, and is Wi-Fi sufficient for our intended use case?*
8. *What additional hardware or infrastructure is required to reliably develop and test the intended system?*

Hardware

Main Computer

- **CPU**: Broadcom BCM2711 quad-core Cortex-A72 at 1.5 Ghz (Raspberry Pi Ltd, 2026).
- **GPU**: Broadcom VideoCore VI at 500 MHz (Broadcom BCM2711, n.d.).
- **RAM**: 4 Gigabytes of LPDDR4 (Hugging Face, n.d.) (Raspberry Pi Ltd, 2026).
- **Storage**: 16 Gigabytes of eMMC flash storage (Hugging Face, n.d.) (Raspberry Pi Ltd, 2026).
- **Operating system**: Custom version of Raspberry Pi OS (Reachy Mini OS) (Pollen Robotics, n.d.).
- **Architecture:** ARM v8 (Raspberry Pi Ltd, 2026).

Camera

- **Camera type**: Raspberry pi camera v3 wide angle (Sony IMX708) (Hugging Face, n.d.).
- **Resolution**: 12 Megapixel (Hugging Face, n.d.).
- **FPS**: 50 FPS video capture (Hugging Face, n.d.).
- **Field of view**: 120 degrees (Hugging Face, n.d.).
- **Is it suitable for pose estimation?**: At the time of writing, we were able to test the Reachy Mini face tracking app. Which did well. There are other people that have used the Raspberry Pi Camera Module v3 for pose detection (Alif, 2023). The camera is the same as on the Reachy Mini.

Audio

- **Microphones**: 4 times the PDM MEMS digital mic (Hugging Face, n.d.).
- **Speaker**: 5W 4Ω speaker (Hugging Face, n.d.).
- **Voice interaction possibility?** We have been able to interact via voice with the Reachy Mini. By using the Reachy Mini conversation app. The Reachy Mini was also able to respond.

Motors & Movement

- **Number of motors**: 9 motors, 2 antenna motors (Dynamixel XL330-M077-T), 6 steward motors (Dynamixel XC330-M288-T) and 1 body turning motor (Dynamixel XC330-M288-PG) (Hugging Face, n.d.).
- **movement**: Figure 1 shows that the head of the Reachy Mini can move upwards and downwards (z-axis), sideways (y-axis) and forwards as well as backwards (x-axis). The antennas on the robot's head are also able to rotate and the robot can rotate its body.

Figure 2 shows that the heads x-axis has a minimum of -1.5cm and a maximum of +2.5cm. The y-axis has a minimum of -4cm and a +4cm maximum. For the z-axis is the minimum -4cm and the maximum +2.5cm. The head roll has a -40 degrees minimum and +40 degrees maximum. The head pitch has the same maximum and minimum as the head roll. The robot body has a -155 degrees minimum and +155 degrees maximum. Both the antennas can be moved 180 degrees both ways (Hugging Face, n.d.).

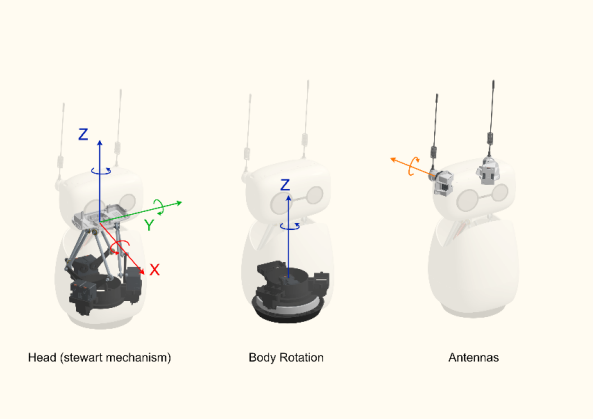
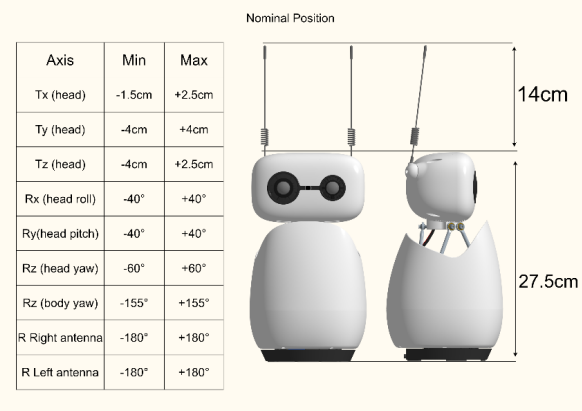


Figure 1: motor movement axis of the Reachy Mini: image by Pollen Robotics



Axis	Min	Max
Tx (head)	-1.5cm	+2.5cm
Ty (head)	-4cm	+4cm
Tz (head)	-4cm	+2.5cm
Rx (head roll)	-40°	+40°
Ry(head pitch)	-40°	+40°
Rz (head yaw)	-60°	+60°
Rz (body yaw)	-155°	+155°
R Right antenna	-180°	+180°
R Left antenna	-180°	+180°

Figure 2: minimum and maximum movement Reachy Mini: image by Pollen Robotics

Other Sensors

- **IMU**: the wireless of the Reachy Mini we use for this project has an inertial measurement unit (Pollen Robotics, n.d.).
- **Position**: All 9 motors have a position sensor (Robotis, n.d.), (Robotis, n.d.), (Robotis, n.d.).

Computing Power and AI models

Local vs. External Processing

There are two possible approaches for processing AI models with the Reachy Mini: local processing and external processing.

With local processing, the AI model runs directly on the Reachy Mini's internal hardware. The Reachy Mini Wireless uses a Raspberry Pi Compute Module 4 with 4 GB of RAM and 16 GB of flash storage (Hugging Face, n.d.). This approach has the advantage that the camera data can be processed directly on the robot, without having to continuously send it to another computer. This can reduce network dependency and communication latency. Another advantage of local AI is that the data does not have to be sent to an external AI service. Local AI generally means that inference is performed on the user's own hardware rather than on a remote server (Local LLM Team, sd). For our application, this could be beneficial because the camera captures a patient's movements.

However, local processing is limited by the hardware available in the Reachy Mini. The robot's CM4 is primarily intended to operate the robot and its applications and does not provide the computing power of a dedicated modern GPU system. Therefore, more demanding AI models may not achieve sufficient performance when running directly on the robot. With external processing, the Reachy Mini can capture the camera data and send it to another computer. The external computer performs the AI inference and sends the resulting pose or movement information back to the Reachy Mini. This allows us to use significantly more powerful hardware for AI processing. The disadvantage is that the system becomes dependent on the network connection and additional latency is introduced. Pollen Robotics has tested the Reachy Mini and an external laptop on the same 5 GHz local network and measured approximately 100 ms total glass-to-glass latency in their media setup (Pollen Robotics, sd). They also note that the actual latency depends on the robot hardware, network and client hardware.

For our project, the most suitable approach therefore depends on whether the selected pose-estimation model can achieve sufficient performance on the Reachy Mini itself.

Requirements of AI Models

The AI component of our application can be divided into two different tasks:

1. Pose estimation: determining the position of a person's body joints.
2. Activity or movement recognition: determining which movement the person is performing and potentially whether it is performed correctly.

ADSAI has already been experimenting with some models:

Model	Type	Description	Computational demand
BlazePose	2D	Fast 2D human pose detection	Relatively low
ViTPose	2D	Modern transformer-based 2D pose estimation	Medium–high
WHAM	3D	3D human motion/pose estimation	High
CameraHMR	3D	Recent 3D human pose and shape estimation	Very high

Tabel 1: Known ADSAI AI models, by Maarten Vaessen

BlazePose is interesting for initial testing because lightweight pose-estimation pipelines can run on Raspberry Pi hardware. Google provides an official MediaPipe Pose Landmarker example specifically designed to run on a Raspberry Pi and perform real-time pose landmark detection from camera images (Pose Landmarker, sd). ViTPose is more computationally demanding. Implementations are available for CPU and GPU inference, but some implementations report real-time performance above 30 FPS specifically on modern NVIDIA GPUs and Apple Silicon (Vitpose, sd). WHAM is a 3D human motion estimation model that reconstructs human motion from video. Compared with BlazePose and ViTPose, which focus on 2D pose estimation, WHAM provides 3D motion information. This makes it relevant for more detailed movement analysis, but also more computationally demanding than the 2D approaches (WHAM, sd). CameraHMR is even more demanding. Its implementation uses PyTorch and CUDA 11.8 in the documented setup, while the project is designed for detailed 3D human pose and shape estimation (CameraHMR, sd). This makes it considerably less suitable as a first candidate for local execution on the Reachy Mini.

But, an important limitation is that none of these four models directly solve the activity-recognition part of our application. BlazePose, ViTPose, WHAM and CameraHMR primarily provide information about the person's pose or motion. A separate activity-recognition or movement-classification step would therefore likely be required. This distinction is important because our intended application requires the system to eventually identify movements and determine whether they are performed correctly.

Expected Limitations

Several factors can limit the ability to run these models directly on the Reachy Mini.

Processing power

- The Reachy Mini Wireless has a Raspberry Pi Compute Module 4 with 4 GB RAM and 16 GB storage (Hugging Face, n.d.). This is significantly more constrained than the hardware commonly used for computationally intensive computer-vision models.

Memory

- Only 4 GB of RAM is available. The operating system, Reachy Mini software, camera processing and AI model would all compete for the available memory.

Model complexity

- The four candidate models have very different computational requirements. A lightweight 2D model such as BlazePose is much more realistic for local testing than a complex 3D human mesh/pose model such as CameraHMR.

Real-time requirements

- Our application is intended to react to a patient's movements. Therefore, simply being able to execute a model is not enough. We need to determine whether it can run with an acceptable frame rate, inference time and response time. For example, a model that processes one frame every several seconds may technically run on the robot but would not be suitable for real-time interaction.

Competing processes

- The Reachy Mini also needs to operate its robot functions, camera, audio and other software. Running an AI model simultaneously may reduce the resources available for the rest of the system.

Network latency

- If processing is moved to an external computer, the network introduces additional latency.

External Processing Alternative

If the Reachy Mini cannot process the selected AI model sufficiently fast, the AI processing can be moved to an external computer. The Reachy Mini supports communication with external computers through its software interfaces, for example GStreamer (GStreamer Installation, sd). The official API provides network connectivity to the robot, while the SDK supports access to robot control and media functionality. In terms of network architecture, the Reachy Mini and external computer could be connected to the same access point, with the access point providing a connection to Eduroam and the internet. You can also use network segmentation to isolate the robot and related devices from the rest of the network so the External Computer can only send and receive data from the Reachy Mini, while the Reachy Mini itself can connect trough Eduroam to the Internet.

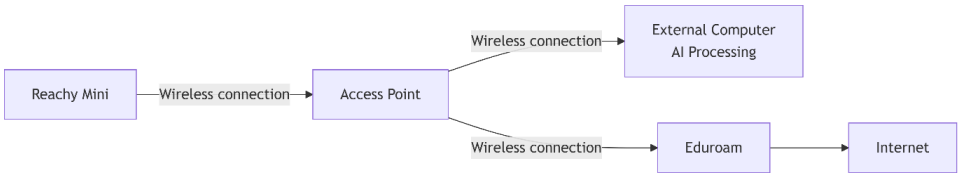


Figure 3: Reachy Mini with external AI computing

This approach would allow us to use more computationally demanding without being restricted to the processing capabilities of the Reachy Mini. The main trade-off is that the system becomes dependent on the external computer and the wireless network connection. The external computer must also be sufficiently powerful to run the selected model in real time. For the first prototype, this architecture is worth testing alongside local processing rather than immediately choosing one approach.

Other models

TSM (TSM-MobileNetV2):

Temporal Shift Module (TSM) is a lightweight mechanism designed to enable efficient temporal modeling. Temporal modeling is the process of using past frames to enhance the understanding of events in a sequence (ScienceDirect, n.d.). In video understanding using 2D convolutional neural network (CNNs), without adding additional computation. Instead of using heavy 3D convolutions for temporal modeling. TSM shifts output from a CNN layer across video frames. This shift allows each frame to receive information from neighboring frames, simulating the effect of 3D convolutions but at the computational cost of a 2D CNN. TSM shifts only a fraction of the output, otherwise data movement would be heavily increased (Lin, Gan, & Han, TSM: Temporal Shift Module for Efficient Video Understanding, 2019).

TSM is licensed under the MIT license. The Github repository contains mostly python code but also provides examples to test and train the model. Figure 4 shows how the model is able to detect the two fingers zoom signal (Lin, Gan, & Han, TSM: Temporal Shift Module for Efficient Video Understanding, 2024).



Figure 4: TSM detecting two fingers zoom: image by (Lin, Gan, & Han, TSM: Temporal Shift Module for Efficient Video Understanding, 2024)

MoViNets (MoViNet-A0):

MoViNets or Mobile Video Networks is a family of efficient video networks process video frames as they arrive. It uses stream buffering technics to process video in not overlapping parts of a fixed length. MoViNets uses temporal ensembling. Which is a technique that recovers some accuracy that is lost when using stream buffers. Without adding any inference cost. Two streaming MoViNets are trained independently with identical architectures but at half the frame rate, keeping the same total duration and thus halving each model's input frames. At inference, the same video is fed into both networks, with one network's frames offset by a single frame

relative to the other. The output from the models is averaged to provide a final prediction (Kondratyuk, et al., 2021).

TensorFlow shows how MoViNet can be setup. Which requires ffmpeg, mediapy, opencv, tensorflow, Tqdm, medipy, numpy matplotlib and PIL. In Figure 5 the MoViNet detects jumping jacks. The model gets more confident in its detection over time (Tensorflow, 2024).

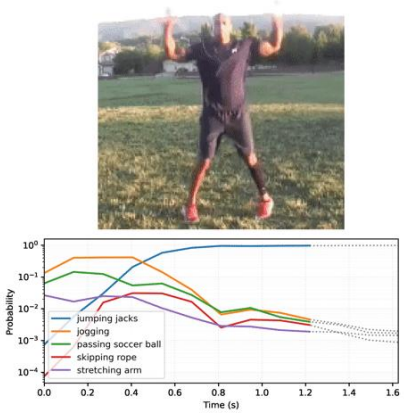


Figure 5: MoViNet jumping jacks detection: image by TensorFlow.

GraphEnet:

GraphEnet uses a graph neural network (GNN). The GNN is made of two components. The initial graph is built from event stream and it uses a method named SCARF (Set of Centre Active Receptive Fields) in parallel to the GNN to process the events to usable data. The model looks at parts of the image and draws a straight line through the moving pixels. If a line isn't clear enough, it is ignored. This drawing process runs in parallel with the GNN. This allows the model to handle large amounts of datapoints every second (Goyal, Thuong, Glover, Mizuno, & Bartolozzi, GraphEnet: Event-driven Human Pose Estimation with a Graph Neural Network, 2025).

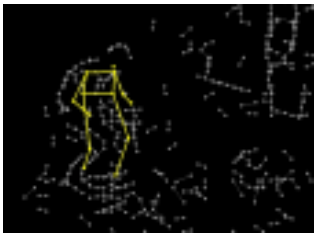


Figure 6: Result GaphEnet: image by (Goyal, Thuong, Glover, Mizuno, & Bartolozzi, GraphEnet: Event-driven Human Pose Estimation with a Graph Neural Network, 2025).

As mentioned GraphEnet uses a graph neural network. However, as visible in Figure 6 it is still represented in skeleton form in the final output.

The model is available on Github under the Attribution-NonCommercial-ShareAlike 4.0 International license. Which allows its use and modification if the following conditions are met: the (original) authors must be properly credited, the derivative work is under the same license and it is not used for commercial purposes. The code is written in python and requires additional libraries like pytorch-geometric as provided in the requirements.txt (Goyal, Thuong, Glover, Mizuno, & Bartolozzi, GraphEnet-NeVi-ICCV2025, 2025).

FSPose

FSPose is a diverse framework designed to balance accuracy and computational needs in human pose estimation for videos. FSPose adapts its architecture based on the type of frame and does not use the same network for all frames like other models. The framework divides video frames into Keyframes and non-keyframes. For keyframes it uses the HRNet W32 neural network. Which has higher accuracy and computational needs than the neural network used for the non-keyframes. Which is the MobileNetV2 which has lower accuracy, but also lower computational needs. When it has processed a keyframe, it will process the following non-keyframes with the lighter neural network. By splitting the frames this way over different neural networks, lowers the average load (Xu, Komorita, & Kawamura, FSPose: A Heterogeneous Framework with Fast and Slow Networks for Human Pose Estimation in Videos, 2023).

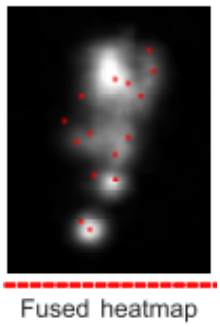


Figure 7: Result FSPose: image by (Xu, Komorita, & Kawamura, FSPose: A Heterogeneous Framework with Fast and Slow Networks for Human Pose Estimation in Videos, 2023).

Figure 7 is an example of the end result provided by FSPose. Which creates a combination of a heatmap with dots.

FSPose is written primarily in python and the Github repository includes instructions to install the model via docker. But also contains the code, including requirements.txt. From the repository we can also conclude that the authors used Cuda to run the model. It is unclear if Cuda is required to run the model. The model is licensed under the Apache license version 2.0 (Xu, Komorita, & Kawamura, FSPose: A Heterogeneous Framework with Fast and Slow Networks for Human Pose Estimation in Videos, 2023).

PySlowFast:

PySlowFast is an Apache 2.0 licensed video understanding codebase from Facebook AI Research (FAIR) that provides video classification models. The repository includes python implementations of multiple methods. The goal of PySlowFast is to provide a high-performance pytorch codebase. Which provides video backbones for video understanding research on different tasks. PySlowFast includes implementations of the following backbone architectures: SlowFast, Slow, Non-local Network, C2D, MViTv1/MViTv2, Rev-ViT/Rev-MViT, I3D and X3D. Figure 8 shows a picture of the SlowFast demo. Which clearly shows the ability of the model to detect what the people are doing (Meta, 2026).

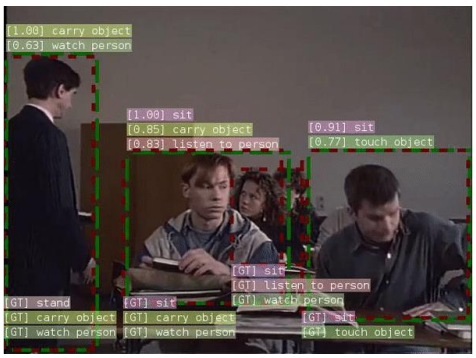


Figure 8: SlowFast demo: image by Meta.

SAM2 (Segment Anything Model 2):

SAM2 (Segment Anything Model 2) is a unified foundation model designed for promptable visual segmentation in both images and videos. SAM2 processes videos frame-by-frame using a streaming architecture equipped with a memory bank. The memory bank stores spatial information and object pointers from previous frames. It maintains a FIFO queue of the most recent frames and prompted frames. Which allows the model to remember object's appearance and trajectory. Once prompted the model predicts the segmentation mask for that frame and propagates it to subsequent frames. If the model loses the object, the user can provide a new prompt in a later frame (Ravi, et al., n.d.). SAM2 is however not feasible to run on the Reachy Mini, because it requires CUDA (Gallagher, 2024).

SAM2 is licensed under the Apache license version 2.0 but also contains cc-torch which is licensed under the 3-clause BSD license. The code is written in python, primarily using Jupyter notebooks. Figure 9 shows how SAM2 is able to detect humans and vehicles (Meta, 2026).

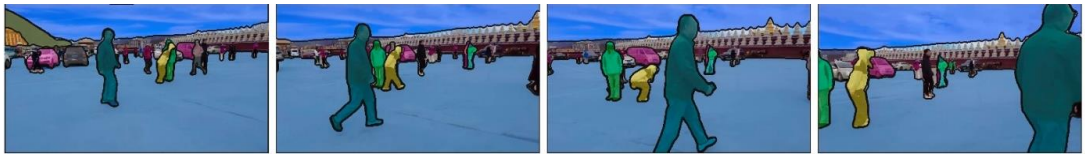
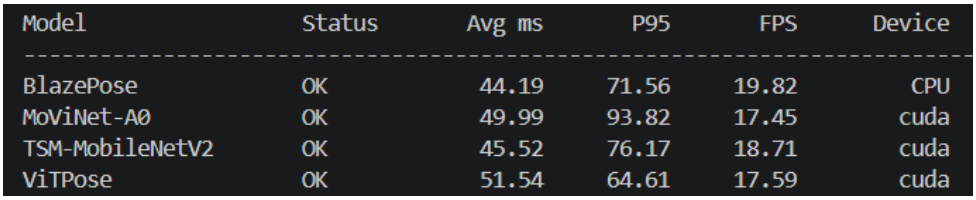


Figure 9: SAM2 detecting vehicles and people: image by Meta.

Selected AI models for benchmarking

Figure 11 shows the benchmark results for: BlazePose, MoViNet-A0, TSM-MobileNetV2 and ViTPose. These benchmarks were run on the laptop of Cas, which has a more powerful CPU and dedicated GPU compared to the Reachy Mini. Also, the RAM of the Reachy Mini is 4 GB, which is lower than the RAM of the Laptop. The benchmark was conducted using a 10 second video, without pose estimation. In the benchmark on the Laptop, the BlazePose model outperforms MoViNet-A0, TSM-MobileNetV2 and ViTPose in speed. Though the BlazePose model was not tested with GPU acceleration in our implementation, it still outperforms the other models which are able to use the GPU. In this benchmark, BlazePose has an average latency of 18.90 ms, a P95 latency of 20.69 ms and an FPS of 52.92.

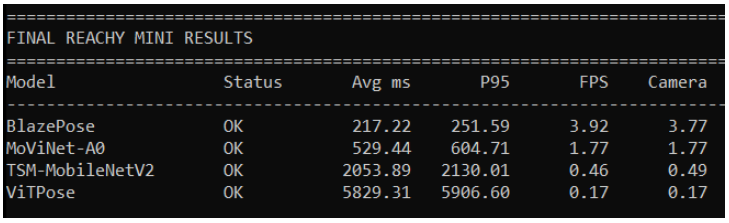


Model	Status	Avg ms	P95	FPS	Device
BlazePose	OK	44.19	71.56	19.82	CPU
MoViNet-A0	OK	49.99	93.82	17.45	cuda
TSM-MobileNetV2	OK	45.52	76.17	18.71	cuda
ViTPose	OK	51.54	64.61	17.59	cuda

Figure 10: Laptop benchmark status

Benchmark	Model	Avg. Latency (ms)	P95 (ms)	FPS	RAM / VRAM (MB)
CPU	BlazePose	18,90	20,69	52,92	178,1
CPU	MoViNet-A0	36,05	45,25	27,74	114,0
CPU	TSM-MobileNetV2	139,28	185,11	7,18	178,6
CPU	ViTPose	197,80	258,12	5,06	70,1
GPU	MoViNet-A0	37,03	54,45	27,01	19,8
GPU	TSM-MobileNetV2	36,51	41,27	27,39	103,0
GPU	ViTPose	48,92	61,54	20,44	350,5

Figure 11: Laptop benchmark results



FINAL REACHY MINI RESULTS

Model	Status	Avg ms	P95	FPS	Camera
BlazePose	OK	217.22	251.59	3.92	3.77
MoViNet-A0	OK	529.44	604.71	1.77	1.77
TSM-MobileNetV2	OK	2053.89	2130.01	0.46	0.49
ViTPose	OK	5829.31	5906.60	0.17	0.17

Figure 12: Benchmark results Reachy Mini

Figure 12 shows the benchmark results on the Reachy Mini. BlazePose performs the best of the tested models, with 3.92 FPS. MoViNet-A0 has a 55% worse FPS rate than BlazePose. TSM-MobileNetV2 has a approximately 88% worse FPS rate and ViTPose has an approximately 96% worse FPS rate than BlazePose. We were able to access the camera on the Reachy Mini directly, which removes the 10 fps cap. Using OpenCV we were able to get approximately 12 fps while

using MoveNet Lightning INT8. Which is a 25% improvement over the previous 10 fps. We were also able to mount an external ssd using the usb-c port on the robot.

We have tried to build models for the Reachy Mini's GPU using OpenGL and Vulkan. We were unable to do this. Because we were unable to disable the compilation with the AES instruction set. We cannot use the robots GPU. So, we tried to run the pose detection on an laptops GPU instead with the camara stream from the Reachy Mini. We tested this and compared this to the newest CPU implementation we made.

Performance Evaluation and Deployment

1. Performance results

Two configurations were tested for the real-time activity recognition system: local processing on the Reachy Mini and remote pose estimation on an Ubuntu laptop equipped with an NVIDIA GeForce GTX 1650 GPU.

Metric	**Local processing on Reachy Mini**	**Remote processing on laptop**
Processing hardware	Reachy Mini CPU	NVIDIA GTX 1650 GPU
Inference runtime	LiteRT / TensorFlow Lite	ONNX Runtime with CUDA
Camera input resolution	192 × 108 pixels	192 × 108 pixels
Measured processing speed	23.80 FPS average	Typically 30–31 FPS
Mean pose inference time	34.32 ms	Approximately 5.1 ms in earlier measurements
Network required for inference	No	Yes
Session and robot control	Local	Remains on the Reachy Mini

Table 2: Comparison local Reachy Minin processing and remote laptop processing.

The local benchmark processed 4,721 frames over 198.40 seconds, achieving an average of 23.80 FPS. Pose inference took an average of 34.32 ms per frame, with a 95th-percentile inference time of 83.62 ms. The session update logic took an average of 2.46 ms.

The laptop implementation typically processed 30–31 frames per second, with occasional drops to between 16 and 28 FPS. ONNX Runtime confirmed that the CUDA execution provider was active, allowing the laptop's GPU to accelerate pose estimation.

The laptop therefore achieved a higher frame rate, but the difference in overall processing speed was relatively small considering the substantial difference in computing power between the two devices.

2. Investigating the frame-rate limitation

The results suggest that the laptop's GPU is not necessarily the limiting factor in the remote configuration. The camera is configured to provide a 192 × 108 pixel MJPEG stream, and the current camera configuration uses a default frame duration of approximately 33.3 ms, corresponding to roughly 30 FPS.

The Reachy Mini's IMX708 camera supports higher frame rates in certain sensor modes. For example, some cropped modes can operate at around 120 FPS, while a full-field-of-view mode can reach approximately 56 FPS. These are sensor-mode limits rather than guaranteed application frame rates.

It would therefore be possible to investigate higher frame rates by changing the camera configuration and adjusting the server's capture loop. However, the actual capture rate would need to be measured independently before concluding that the camera is the only bottleneck. JPEG encoding, network transmission, frame decoding and other application overhead may also affect the remote pipeline.

More importantly, the current results do not indicate that higher FPS is necessary for a satisfactory user experience. Both configurations already operate at real-time speeds.

3. Activity recognition and user experience

The exercise-session tests demonstrated that the system can provide an interactive, guided workout experience. The Reachy Mini speaks to the user, introduces exercises, demonstrates movements using its head and antennas, tracks the user's body and provides feedback during the session.

The local benchmark successfully counted all four repetitions in the squat and side-bend rounds. The laptop implementation generally maintained a higher frame rate, but this does not automatically translate into better activity recognition.

4. Preferred deployment approach

Although the laptop has significantly more computing power, the Reachy Mini achieved an average of 23.80 FPS while running MoveNet locally on its CPU. This is already a strong result for an embedded device and is sufficient to support responsive exercise interactions. Running the complete application locally on the Reachy Mini is therefore the preferred deployment approach, provided that activity recognition and repetition counting are sufficiently reliable. Local processing reduces dependence on an external laptop and network connection, simplifies deployment and keeps the system self-contained. Remote inference remains useful during development, experimentation and model optimization. It provides additional computing headroom and allows alternative models to be evaluated without being constrained by the robot's CPU.

5. Conclusion Evaluation and Deployment Considerations

The performance evaluation shows that local inference on the Reachy Mini achieves 23.80 FPS on average which is more than enough for activity recognition.

Higher camera frame rates could be investigated by modifying the camera configuration and capture pipeline, but this should be treated as an optimization rather than a requirement. The current frame rates are already suitable for real-time interaction.

Ultimately, the goal is not to maximize FPS, but to provide reliable movement recognition and a smooth, engaging exercise experience.

Connectivity & Networking

Available Connections

- **Wi-Fi**: Wi-Fi 5 with 2.4Ghz and 5Ghz dual-band (Hugging Face, n.d.).
- **Bluetooth**: Version 5.0 with Bluetooth low energy (BLE) (Raspberry Pi Ltd, 2026).
- **USB**: The Reachy Mini wireless has a USB type-c port for data transfer (Hugging Face, n.d.).
- **Ethernet?:** The Reachy Mini does not have an Ethernet port, even do the main board has a chip to support Gigabyte ethernet (Raspberry Pi Ltd, 2026) (Hugging Face, n.d.).
- **Other interfaces?**: the reachy mini also has a charge port. It cannot be charged with a USB type-c cable (Hugging Face, n.d.).

How Does Reachy Connect?

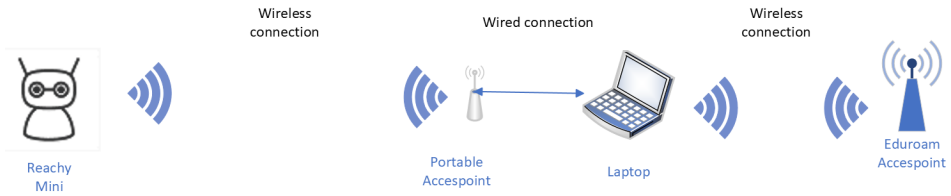


Diagram 3: Reachy Mini to Eduroam network

In diagram 1 we show how the Reachy Mini can connect to the internet. A portable accesspoint is used to share the laptops Eduroam connection with the Reachy Mini. The wireless accespoint creates its own network which allows us to connect to the Reachy Mini with a stable connection.

Is Wi-Fi Enough?

- **bandwidth**: The WI-FI bandwidth for the 5Ghz band is up to 433mbps (Infineon, n.d.) (Raspberry Pi Ltd, 2026). If the camera streams its footage to a laptop this is not expected to consume more than 24 mbps of bandwidth (Hakopyan, 2025). Which means that there is still enough bandwidth for the other functionality that is needed for the project.
- **Reliability**: We have tested the Reachy Mini while connected to the portable accespoint. From when we started using this portable accespoint the connection has been stable while testing apps like the Face-tracking app.

Software & Available Functions

Robot Software / SDK

There are two SDKs available for the Reachy Mini. One python SDK and one JavaScript SDK. According to the documentation both SDKs provide full control of the robot (Hugging Face, n.d.). The Reachy Mini python SDK is supported on Windows, MacOS and Linux. The python SDK requires python version 3.10-3.12 and the latest version of git, including git LFS. The reachy-mini python package needs to be installed into a virtual environment which can then be used to program the robot (Huggingface, n.d.). The Reachy Mini ships a JavaScript browser SDK over WebRTC. The npm package pollen-robotics/reachy-mini-sdk exposes: the ReachyMini class, the host shell and embed adapter. The documentation for the Reachy Mini JavaScript SDK also shows that WebRTC, audio data and video data are peer to peer with the robot and the controlling device. This is not the case for the http signals, they go through a signaling server and then to the robot (Hugging Face, n.d.).

Available APIs

The Swager UI of the Reachy Mini is reachable from: http://reachy-mini.local:8000/docs. In the Swager UI all available API options are listed. Through the API we are able to Install, remove and manage apps. We are also able to get a Unified robot description (URDF) through the API, move the robot into different positions, play sounds and more. (Hugging face, n.d.)

Hardware Access

The hardware can be accessed using the python SDK with the following functions. We can record audio with the Reachy Mini by using the function: mini.media.start_recording() and mini.media.get_audio_sample() to save the audio sample. To play recorded audio on the Reachy Mini the mini.media.push_audio_sample() function can be used. The frames from the camera can be accessed using mini.media.get_frame() when the media backend is set. The Reachy Mini can be moved using the mini.goto_target() function. In this function will then be specified what part the Reachy Mini needs to move and how long that will take. So for example, the body can be turned 20 degrees using body_yaw=np.deg2rad(20) and can the antennas be moved 50 degrees using antennas=np.deg2rad([50, 50]) (Hugging Face, n.d.).

Available AI Integration

The Reachy Mini supports head tracking by using the mini.start_head_tracking(). Function (Hugging Face, n.d.). While this is not sufficient for full pose estimation it might be useful to detect the user before the session has started.

Conclusion & Recommendation

The main question: to what extent can the Reachy Mini detect a person, recognize physiotherapy-related exercises, and provide expressive feedback within its computational constraints? Can be answered positively: all are possible on the robot itself.

The Reachy Mini Wireless is built around a Raspberry Pi Compute Module 4 , with a 12 MP wide-angle IMX708 camera, four microphones, a 5 W speaker, an IMU and nine Dynamixel motors. All hardware is accessible via the Python SDK, the JavaScript SDK and the REST API, covering camera frames, audio recording and playback, motor control and built-in head tracking. The software ecosystem is therefore fully sufficient for the intended application.

Computationally, the robot is constrained but still usable. The GPU of the Pi Compute Module 4 proved unusable so AI must run on the CPU. Heavy models are unsuitable locally. The lightweight model MoveNet achieved 23.80 FPS locally, which is sufficient for real-time interaction. Direct camera access removed the default 10 FPS cap.

Crucially, pose estimation alone does not classify exercises. A separate recognition step is required, and tests showed this works: the local pipeline successfully counted all repetitions during squat and side-bend rounds while the robot guided the session with speech, head movements and antenna movement.

External processing is feasible but unnecessary for deployment. A laptop with a GTX 1650 reached 30-31 FPS remotely, yet the gain was marginal because computing power is not the bottleneck. Wi-Fi is adequate: the stream uses well under 24 Mbps of the available 433 Mbps, and connections were stable. A portable access point sharing an Eduroam connection solves campus network restrictions. Required additional equipment is limited to a development laptop, a portable access point.

Recommendations

- Deploy locally. The 23.80 FPS local pipeline is sufficient and keeps the robot self-contained. Which avoids dependence on a laptop in a patient's home
- Use MoveNet as the local pose estimator. Which provided the best performance on the Reachy Mini's CPU.
- Focus follow-up research on CMAS exercises and clinical robustness. The technical feasibility is established. The remaining questions are functional. Follow-up research should evaluate the system specifically against the CMAS exercise set, testing how reliably it recognizes correct versus incorrect execution of each CMAS exercise across different patients, body types and lighting conditions.

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D Requirements Verification

M-ADSAI-ZM002-26

Requirements Verification

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Functional Requirements Evaluation

Table 1 – Perception and local processing functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-01	Person Detection	The system shall detect a person within the Reachy Mini's camera view within 5 seconds after the person enters the camera view during an exercise session.	Yes
Explanation:		The system detects a person within the Reachy Mini's camera view within 5 seconds after the person enters the camera view during an exercise session.	
FR-02	Local Processing	The system shall process captured camera data entirely on the Reachy Mini or the connected local computer, without requiring external processing for core activity-tracking functionality.	Yes
Explanation:		The captured camera data is processed on the Reachy Mini Robot and does therefore not leave the robot.	
FR-03	Person Tracking	The system shall track the detected person continuously while the person remains visible within the camera's field of view during an exercise session.	Yes
Explanation:		The robot tracks the user while the exercises are performed.	
FR-05	Pose Estimation	The system shall provide the input representation required by the selected activity recognition approach, which may be an explicit pose/skeletal representation or image/video data.	Yes
Explanation:		The system provides the input representation required by the selected activity recognition, using skeletal representation.	
FR-18	User Detection Loss	The system shall detect that the user is no longer visible to the camera within 2 seconds after the user leaves the camera's field of view during an exercise session.	Yes
Explanation:		The system detects that the user is no longer visible to the camera within 2 seconds after the user leaves the camera's field of view during an exercise session.	

Table 2 – Robot behaviour and interaction functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-04	User Following	The system shall move the Reachy Mini's head to follow the detected user while the user remains within the camera's field of view during an exercise session.	Yes
Explanation:		The Reachy Mini robot follows the user for the exercises by moving its head.	
FR-12	Audio Feedback	The system shall provide audio feedback through the Reachy Mini's speaker whenever an audio feedback response is triggered.	Yes
Explanation:		The robot provides audio feedback when an exercise is performed.	
FR-13	Head Movement Feedback	The system shall use movements of the Reachy Mini's head to provide feedback to the user whenever a head-movement feedback response is triggered.	Yes
Explanation:		If an exercise is performed successfully the Reachy Mini will do a short dance with its head and antennas.	
FR-14	Antenna Movement Feedback	The system shall use movements of the Reachy Mini's antennas to provide feedback to the user whenever an antenna-movement feedback response is triggered.	Yes
Explanation:		If an exercise is performed successfully the Reachy Mini will do a short dance with its head and antennas.	

Table 3 – Activity and exercise recognition functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-06	Static Movement Recognition	The system shall recognize all static positions listed as supported activities during an exercise session.	Yes
Explanation:		The system is able to detect how long the user performs the arms up exercise.	
FR-07	Dynamic Movement Recognition	The system shall recognize all dynamic movements listed as supported activities during an exercise session.	Yes
Explanation:		The system can recognize chair rise and pick-up exercises.	
FR-08	Physiotherapy Exercise Recognition	The system shall recognize all physiotherapy exercises listed as supported exercises during an exercise session.	Yes
Explanation:		The system can recognize chair rise, pick-up and arms up exercises.	

Table 4 – Assesment and feedback functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-09	Movement Assessment	The system shall assess the execution of supported physiotherapy exercises using the predefined movement criteria for each exercise.	Yes
Explanation:		The system is able to detect length of an exercise (arms up) and if an exercise is performed (Chair rise and pick-up).	
FR-10	Repetition Detection	The system shall detect completed repetitions according to the predefined repetition criteria during an exercise session.	Yes
Explanation:		The system can recognize chair rise, pick-up and arms up exercises in repetitions.	

Table 5 – Session management functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-11	Session Management	The system shall allow the user to start and end an exercise session.	Yes
Explanation:		The system allows the user to start and end an exercise session.	
FR-12	Exercise Monitoring	The system shall monitor the user's movements continuously throughout an active exercise session.	Yes
Explanation:		The system monitors the user throughout the exercise session.	
FR-13	Exercise Result	The system shall provide the user with the detected exercise results and assessment results at the end of an exercise session.	Yes
Explanation:		The Reachy Mini tells the user, the results after an exercise session and when an exercise is detected.	

Non-Functional Requirements Evaluation

Table 6 – Non-Functional Requirements

ID	Title	Requirement	Requirement met by artifact
NFR-01	Privacy	The system shall ensure that captured video and session data remain on the Reachy Mini and/or the connected local computer for the entire duration of system operation and shall not be transmitted to external systems.	Yes
Explanation:		The video and session data stays on the Reachy Mini	
NFR-02	Recognition Accuracy	The system shall provide activity recognition for the supported activities when evaluated using the predefined test dataset. The recognition performance shall be measured and reported.	Yes
Explanation:		The system provides activity recognition for the supported activities when evaluated using the predefined test dataset. The recognition performance is measured and shown in figure 1. Class | Full body | Waist-up none | 0.902 | 0.855 wave | 0.918 | 0.873 squat | 0.922 | 0.803 clapping | 0.898 | 0.899 jumping_jacks | 0.887 | 0.861 pushup | 0.381 | 0.137 *Figure 1: per-fold F1 score*	
NFR-03	Reliability	The system shall complete an exercise session without requiring an application restart under normal operating conditions.	Yes
Explanation:		The system completes an exercise session without requiring an application restart under normal operating conditions.	
NFR-04	Usability	The system shall allow first-time users to start an exercise session without assistance from the development team.	Yes
Explanation:		The system allows first-time users to start the session without assistance.	
NFR-05	Safety	The system shall not perform unexpected or uncontrolled robot movements during normal operation.	Yes
Explanation:		The Robot does not perform unexpected or uncontrolled robot movements during normal operation.	

Conclusion

Requirements Verification has shown that all functional and non-functional requirements identified in the SRS have been satisfied in the Reachy Mini prototype. The system has the capability to detect and track users, recognize physiotherapy exercises that are supported, monitor exercise execution and give feedback via audio and robot gestures. Also, the prototype has satisfied the requirements for processing locally, privacy protection, reliability, usability and safety. The recognition accuracy has been tested with the use of a predefined test dataset.

However, despite the fact that all the requirements have been met, there is still need for additional testing and development to make sure that the system will work properly in a healthcare setting. The recognition accuracy should be tested on a greater number of patients and different environments. Generally, the prototype is a good basis for further development into a system which can conduct CMAS assessments at home.